

1. Which of the following devices on the Board is NOT interfaced using I2C?
 - a. Barometer
 - b. Gyroscope
 - c. Bluetooth module
 - d. NFC module

2. In I2C, if the master does not acknowledge the byte, the slave will
 - a. not transmit any further bytes in the current transaction
 - b. re-transmit the byte only if it is explicitly configured to do so
 - c. re-transmit the byte only if it is in transmitter mode
 - d. always re-transmit the byte

3. Which of the following is NOT considered a limitation/disadvantage of I2C?
 - a. Pin Count
 - b. Data Rate
 - c. Transmission Distance
 - d. Address Clashes

4. Which of the following bits is NOT present in I2C_ISR?
 - a. TCR
 - b. TC
 - c. TXE
 - d. NACK

5. STM32 I2C is interfaced to an accelerometer (7-bit address 0x32) which measures the acceleration values in X, Y and Z coordinates as 16-bit values (8 bits stored in L and 8 bits stored in H). The 8-bit register addresses for accelerometer are as follows:
Acceleration at X-axis: ACC_XL (0x10), ACC_XH (0x11)
Acceleration at Y-axis: ACC_YL (0x12), ACC_YH (0x13)
Acceleration at Z-axis: ACC_ZL (0x14), ACC_ZH (0x15)
STM32 does an I2C transaction to read the 16-bit acceleration values in X, Y and Z axes. Assuming autoincrement is enabled by default in the accelerometer and the I2C transaction is successful, list the number of:
 - a) start conditions
 - b) stop conditions
 - c) acknowledges
 - d) not acknowledges
 - e) bytes from master to slave (including slave address)
 - f) bytes from slave to master